



Rivers Never Dry

By Sohyun Jun

“How do we understand a river that did not simply disappear, but was translated into something else while remaining?”

<https://vimeo.com/1224707820>

Introduction

Rivers Never Dry is a data-driven installation that renders the buried waterways that used to flow into the Thames through their physical traces and real-time public data.

Concept and Background Research

London's rapid urbanisation during the Industrial Revolution saw 18th- and 19th-century sewage discharge turns its rivers into open sewers, spreading disease and triggering recurring health crises. In response, many tributaries feeding the Thames were culverted. Although they appear dry today, they still flow beneath our feet as sewers. Their presence lingers nonetheless: parks on land too damp for habitation, willow and alder trees marking former channels, buildings that swerve around buried riverbeds, and Victorian stink pipes emerging awkwardly between roads and buildings (Newman, 2016). Tracing these fragments led to this project's use of public data. Using real-time data on Storm Discharge status from Thames Water, and water quality values from the Environment Agency, the installation reacts to highlight the presence of fragmented traces of lost rivers. It brings the moment when the subterranean rivers emerged into a broader ecological setting and were translated into data.

The collaboration with machines to sense the non-human, such as the river, and the trees and land around it, gives us the opportunity to understand and rethink the way human supremacy has been working in our current world, and its resilience against it.

Technical Implementation

My first iteration involved connecting three motors via vertically running pipe structures to create a single kinetic sculpture. The intention was to symbolise environmental pollution caused by human activity, alongside the river's self-regulating capacity in response to it. This would be expressed through a sculpture that changed continuously depending on the viewing angle, driven by the individual movements of motors controlled by real-time environmental data (Figure 1.1).

I first built a motor assembly through prototyping to test its feasibility. The movement of a single module was based on a simple logic: the section below the motor remained fixed, whilst the section above rotated with the motor shaft. To achieve this, I designed custom 3D-printed brackets and small components: a bearing was fitted to the motor shaft so that the pipe in contact with the bearing stayed fixed, whilst the pipe attached directly to the shaft was free to rotate (Figure 1.2). Building on this principle of stacking vertically while keeping some parts fixed and others moving, I used a layered structure secured with threaded rods and nuts, which allowed the spacing between layers to be adjusted flexibly (Figure 2.1, 2.2).

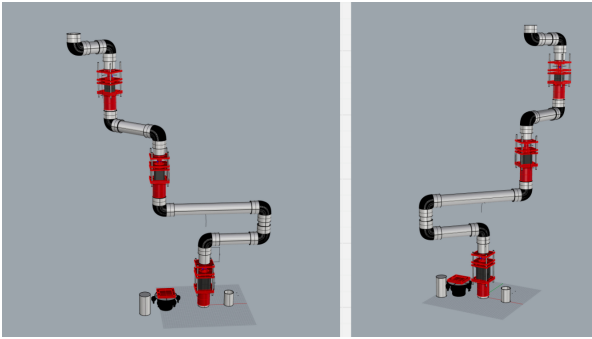


Figure 1.1. 3D modelling of a full kinetic sculpture with three motors

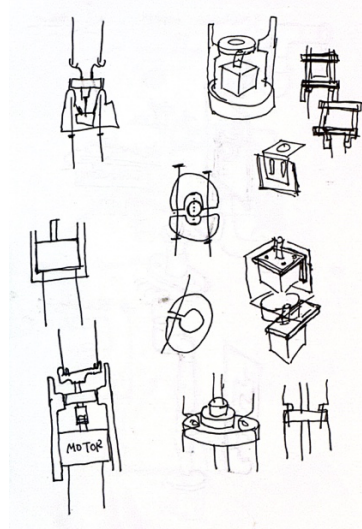


Figure 1.2. Sketches for custom motor structure

However, once the aluminium pipes and metal fittings arrived, discussions with a technician confirmed that the motor would be unable to support the weight of the full assembly (Figure 2.3). The structure alone weighed nearly 20 kg, and the torque required (calculated using the formula " $\text{Torque} = \text{Distance} \times \text{Force}$ ") came out far beyond what my budget and resources could accommodate. This led to a major shift in plans: moving from an entire kinetic to a largely fixed form, using the materials I already had, while finding another way to preserve the work's kinetic character.

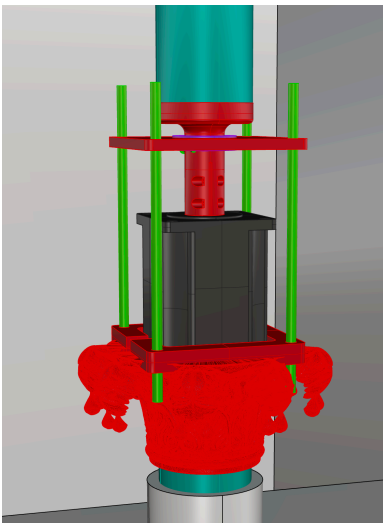


Figure 2.1. Custom motor structure

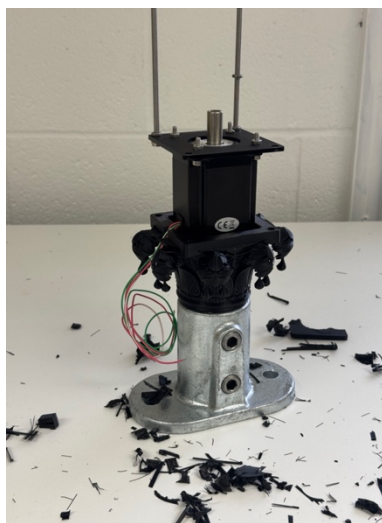


Figure 2.2. 3D prints and assembly



Figure 2.3. Pipe materials

I addressed this by replacing the kinetic expression of real-time data with a digital screen. Early tests of TouchDesigner's fluid simulation using *SPH* and *Sphere* tools (Pelz, 2026) on a standard 27-inch monitor failed to convey the flow through the pipes convincingly; after several rounds of testing, I sourced unconventional long, narrow

monitors, could be mounted it onto the pipe structure, and created a visual in which the river exists as data, its interior visible as though through the monitors. Compared with the physical installation, the software side was relatively straightforward. A Node.js server fetched data every 15 minutes from Thames Water and the Environment Agency (EA), matching the EA's water-quality update interval. The Thames Water data, by contrast, was highly irregular and binary in nature, updating only when a discharge occurred at one of the 573 storm discharge points. To bring this data source into the piece, I mapped the discharge dates, collected manually from December last year to August this year, onto the exhibition's opening hours of 11am to 8pm, generating a 10-minute event during which a discharge appears to occur. During this window, the actual sounds recorded inside the pipes, provided by Thames Water, play on a loop, and visitors are invited to return at the designated times to witness it.

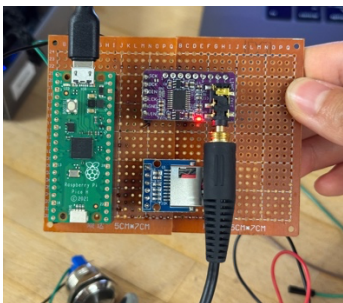


Figure 3.1. Early prototype of Digital to Analog connection

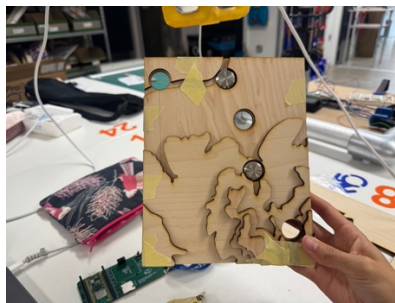


Figure 3.2. Fabrication process of interactive map

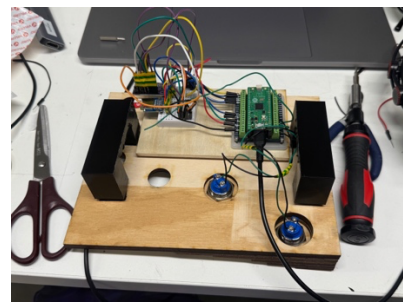


Figure 3.3. Backside of interactive map

In addition to the main piece, I created an interactive map featuring a headset and push buttons. This piece serves as a journal and personal record, captured through my own recorded voice, of the terrain and traces of the river I experienced whilst walking the entire length of the River Effra, from its source to its mouth, as well as visiting the data sensing points along the River Thames. It consists of a Raspberry Pi Pico, a DAC module and a 3.5mm jack module, mounted openly on the reverse of a map visualising the contour lines, rather than concealed within an enclosure (Figure 3.1, 3.2, 3.3).

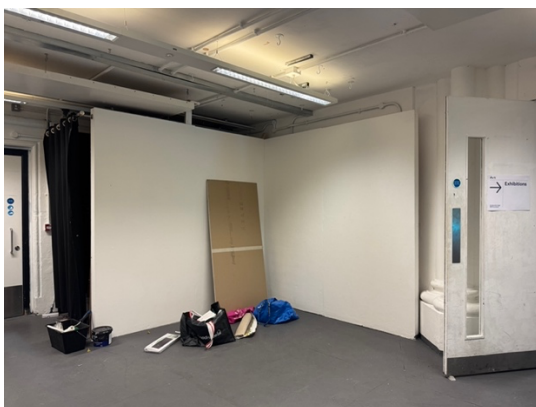


Figure 4.1. Assigned corner for installation

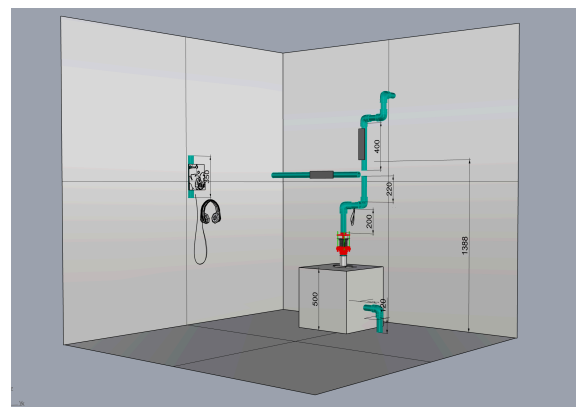


Figure 4.2. 3D Simulation of installation considering site specific features

Reflection and Future Development

As this project utilises physical, bulky pipes as its primary material, it is subject to certain spatial constraints. For this exhibition, the works were primarily mounted on walls; however, should the opportunity arise to exhibit in a different space, it might be possible to adapt them into freestanding sculptures, using the pipes' flexibility in assembly.

In terms of how data is applied to the work, I used both official data provided by organisations and empirical data collected first-hand. In the next iteration, I would like to incorporate data gathered either through self-made sensing devices or by participating in a citizen science group to collect data directly from the river, as there are doubts regarding the validity of the data provided by Thames Water and the EA. The EA, for example, has been accused of cancelling tests and downgrading pollution incidents without investigation (Mooney and Plimmer, 2026). Furthermore, it is only natural to view the data published by Thames Water itself, the organisation responsible for discharging sewage into the river without clear justification, with a critical eye. I would like to develop this project by making more active use of such democratic approaches and the methods of 'Sensing Practice' (Gabrys and Pritchard, 2018).

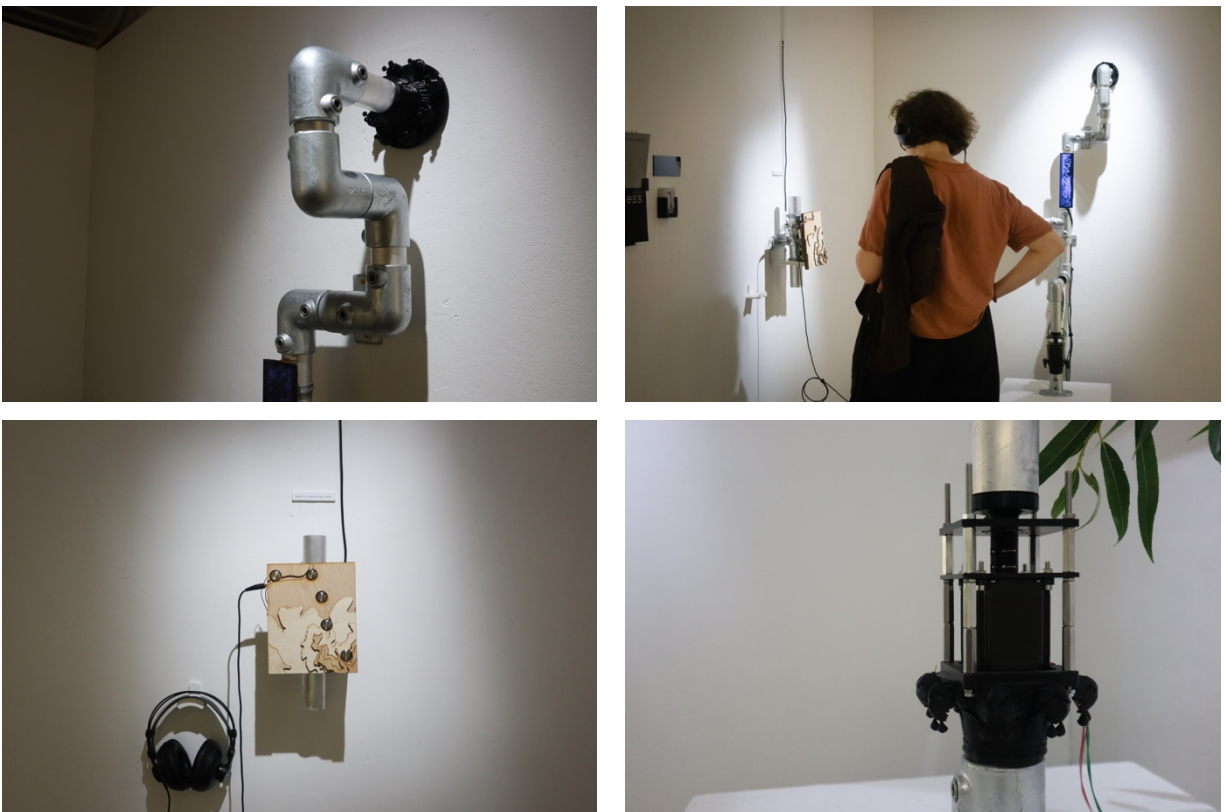


Figure 5. Moments from the exhibition

References

Gabrys, J. and Pritchard, H. (2018) 'Sensing practices', in Braidotti, R. and Hlavajova, M. (eds.) *Posthuman Glossary*. London: Bloomsbury, pp. 394–396.

Mooney, A. and Plimmer, G. (2026) 'The citizen scientists tracking England's polluted waterways', *Financial Times*, 7 September. Available at: <https://www.ft.com/content/a5e58e66-25c1-449f-a0bf-ca9a18b67307> (Accessed: 8 September 2026).

Newman, J. (2016) *River Effra: South London's Secret Spine*. London: Signal Books.

Pelz, J. (2026) *Interactive Water Simulation in TouchDesigner | Tutorial*. Available at: <https://www.youtube.com/watch?v=kcksGqUVJw4> (Accessed: 14 July 2026).

GitHub repository

<https://github.com/devsohyun/rivers-never-dry-degree-show>

AI Declaration

I acknowledge the use of [1] Claude (<https://claude.ai/>) [2] to consult on and refine the system architecture and hardware wiring for my motor setup. On 6 July 2026 I submitted a question to [3] "I have three motors: Nema 23 Hollow Shaft, Nema 17 Hollow Shaft, Nema 14 usual shaft. I would like to connect them to arduino mega (check if using this microcontroller is best). I will use a computer to create a node.js server to request api from Thames water and EA, using this value, process them, then use it to determine how much rotation the stepper motors should take. Show me the main network diagram you suggest, also the hardware plans for arduino pin locations (how to connect three motors and three motor driver), and if I need some kind of special power adapter considering Voltage". [4] The output was then modified further to better align with my own intentions and project structure.

I acknowledge the use of [1] Claude Code in my terminal [2] to generate codes for Node.js server by vibe coding. Between 20 July 2026 and 8 August 2026 I submitted original python codes (https://github.com/devsohyun/discharge-status-visualizer/blob/main/thames_water_discharge.py) with the instruction to [3] "How do I use this python codes in nodejs to build a server that request API every 15 minutes, and send the values via OSC". [4] The output was then modified further to improve its functionality and integration to work with TouchDesigner.

I acknowledge the use of [1] Claude (<https://claude.ai/>) [2] to generate codes for Arduino connection for motor rotation whenever there is serial communication from the server. On 13 July 2026 I submitted the instruction to [3] "How do I make motor rotate when nodejs server send signal via serial with a certain degree set by variable, then rotate back to initial position after 10 minutes". [4] The output was then modified further to improve its functionality and align it with the intentions of the project.